



Opmaterial
海川新材

OCEANPOWER

NEW MATERIAL TECHNOLOGY CO.,LTD

SHENZHEN

JIANGSU

FAP REBAR

GFRP

BFRP

CFRP

Jiangsu Oceanpower New Material Technology Co.,Ltd

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■ OCEANPOWER GROUP



FAP— A Designable Future

Opmaterial
海川新材

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DIRECTORY



Enterprise Profile

Jiangsu Oceanpower New Material Technology Co., Ltd was established in 2016, a wholly-owned subsidiary corporation of Shenzhen Oceanpower New Material Co., Ltd, which was founded in 1998.

Shenzhen Oceanpower is a high-tech company which includes R&D, production, sales, design and service. We offer a wide range of products that include FRP related products such as FRP Rebar (Rockbolt, Dowel), FBG (Fiber Bragg Grating), Colored Asphalt, Industrial additives such as anti-rut agent, modifying agent etc.

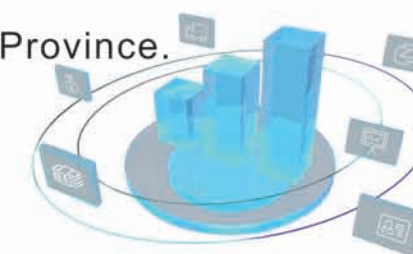


Enterprise Profile



Jiangsu Oceanpower is the FRP specialist in the family, our product line: RoadPower is compliant with the highest domestic and foreign national industrial standard, and have been successfully implemented in many scenarios such as coal-mine, foundation pit, slope protection, highway/bridge construction and subway tunnels; earning us a world-wide reputation for quality and professional service.

We are located in Jurong City, Jiangsu Province.
We humbly welcome you to visit us!



Honor&Certification



■ High-Tech Enterprise



■ Postdoctoral Programme



Honor&Certification



■ Subcommittee 1 on composites and reinforcement fibers in civil engineering of national technical committee 39 of fiber reinforced plastics of standardization administration of china



■ Post-Doctoral Workstation Industrialization Production Base



■ Standards Committee Demonstration Production Base



■ Oceanpower Test Lab



■ China Well-known Trademark



■ TUV Certificate;



■ CE Certificate (FRP Product)



■ CE Certificate (FRP Product)



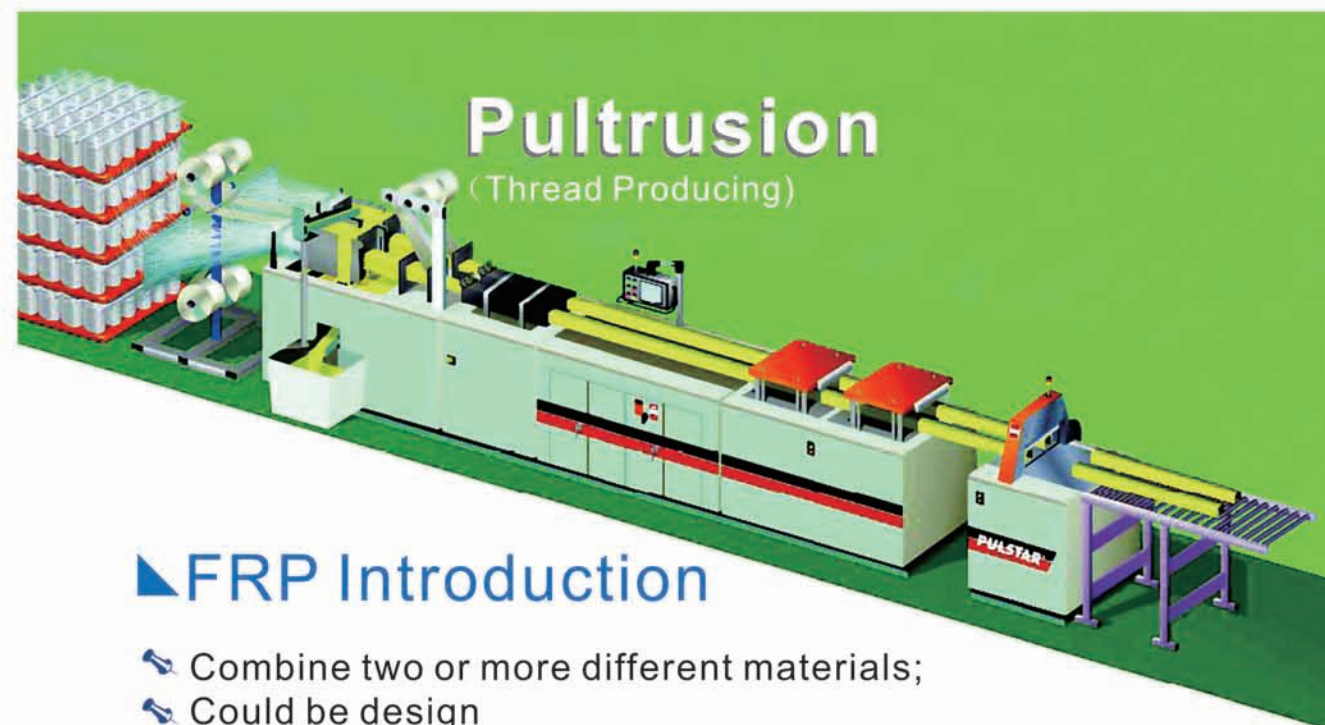
■ ISO9001 Quality Management System



■ ISO14001 Environment Management System



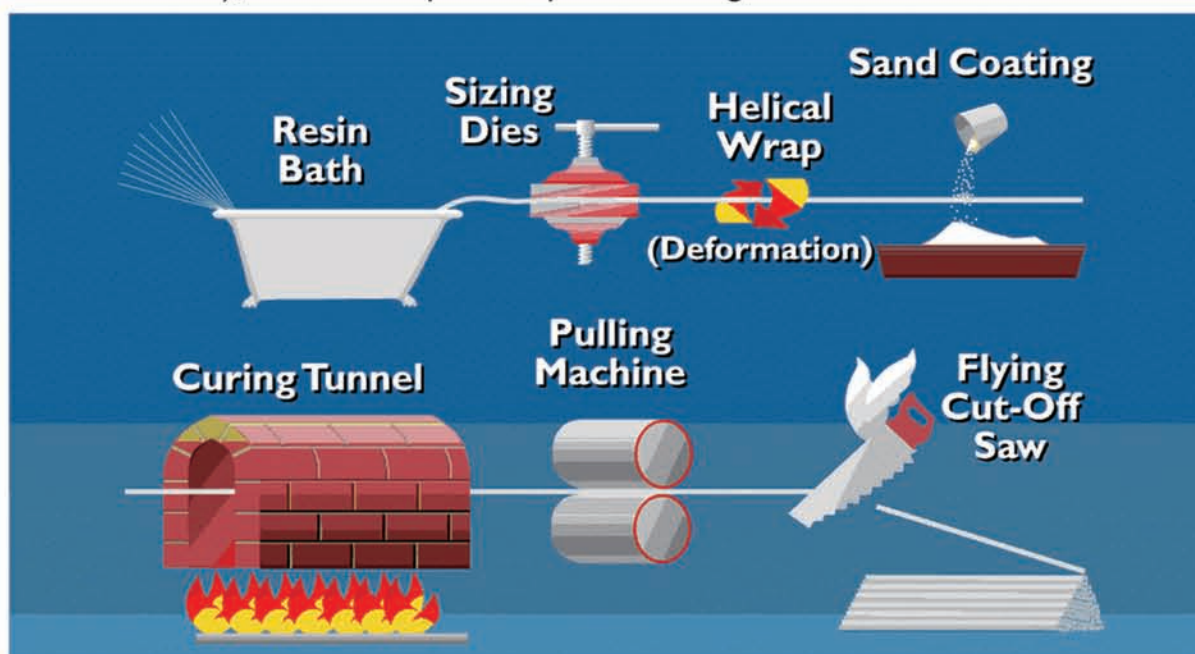
■ ISO18001 Occupational Health and Safety Management Systems



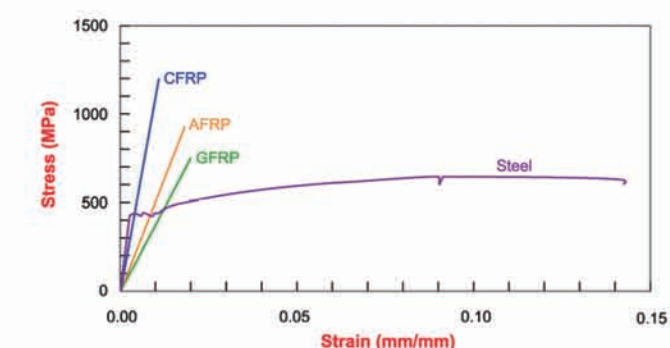
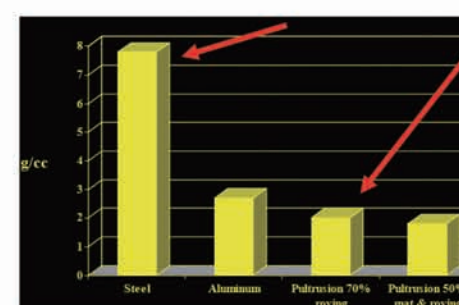
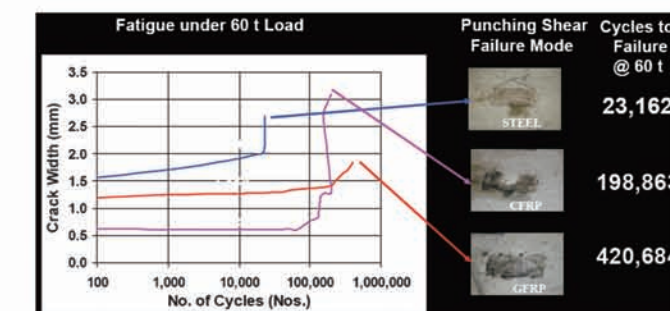
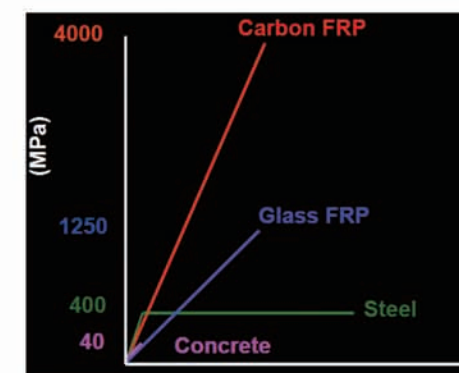
FRP Introduction

- Combine two or more different materials;
- Could be design
- Through composite method that get more strength

Fiber reinforcement polymer, also called Fiber Reinforce Plastic, is a kind composite material which matrix is resin (epoxy, vinylester and unsaturated polyester), fiber as reinforcement (glass, basalt and carbon fiber), after composite processing.



FRP Introduction



GFRP Advantage

High Corrosion Resistance: The base materials used by the bolt are durable materials, and they are molded through the composite process. The life span is up to 100 years. They can be used as permanent support materials

High Tensile Strength: The load is approximately double that of a steel bar with the same diameter

Low Weight: The weight is only 1/4 that of a steel bar with the same diameter. Therefore, the labor intensity is greatly reduced, and the transportation cost is reduced at the same time;

Anti-static: It has no electrical conductivity, and no sparks will be produced when being cut. It is particularly suitable for high gas zones;

Cutability: It avoids damages to cutter heads, and does not delay excavation

Save cost: Use this material as reinforcing bars for road and bridge, could reduce Secondary repair cost.



FRP INTRODUCTION

FRP Rebar	Tensile Strength (MPa)	E-modulus (GPa)	Shear Strength (MPa)	Elongation (%)	Yield Strength (MPa)
GFRP	>600	>40	100	1.5	/
BFRP	>800	>50	100	1.5	/
CFRP	>1800	>120	>100	1.5	/
HRB335	335	200	200~230	16	335
Remark	Dia 4-40mm, more dia could custom-made				

1 FRP, refers to Fiber Reinforced Polymer(Plastic). Existing include G(lass)FRP/ B(asalt)-FRP/C(arbon)FRP/A(ramid)FRP and so on. GFRP is common known as the Glass Fiber Reinforced Polymer(Plastic).

2

FRP is a new high performance material which made of reinforced fiber with matrix resin by certain proportion formation. It's could achieve features of various kinds of applications requirement with different resin. Matrix resin include Unsaturated Polyester, Epoxy, Vinyl-ester and Polyurethane resin etc.

3 Originally, invention of FRP is in order to solve the problem of steel rebar corrosion. As tech development, FRP materials derived a wide variety of products, could used in mining roadway, transit, coastal, municipal construction and other applications.

■ Basalt Fiber



■ Carbon Fiber



■ Glass Fiber





Ordinary steel rockbolt system is widely applied in coal mine tunnel support, these systems mostly for the temporary support measures.

All-thread GFRP rockbolt series product not only save the cost of support project, but also improve the safety and maneuverability

Test Item				MGSL18	MGSL20	MGSL22	MGSL24	MGSL27
Rod	Appearance			Uniform texture, no bubble, no crack				
	Diameter (mm)			18	20	22	24	27
	Tensile Load (kN)			130	160	190	230	290
	Tensile Strength (MPa)			500	500	500	500	500
	Shear Strength (MPa)			100	100	100	100	100
	Tourque (Nm)			45	55	70	100	150
	Anti-static (Ω)			3×10 ⁷	3×10 ⁷	3×10 ⁷	3×10 ⁷	3×10 ⁷
	Flame Resistance	Flaming	Six total (s)	≤25	≤25	≤25	≤25	≤25
			Single (s)	≤6	≤6	≤6	≤6	≤6
		Flameless Combustion	Six total (s)	≤100	≤100	≤100	≤100	≤100
			Single max (s)	≤20	≤20	≤20	≤20	≤20
Anchoring Force (kN)			65	75	85	100	110	
Nut&Tray	Tray&Nut Load			60	70	80	90	100

The advantages of GFRP rockbolt applied in Mining&Tunnel support

- Equal strength all thread
- High tensile strength
- Cuttability
- High load strength
- Low weight, easy to handle and flexibility.

	Unit	Solid Rockbolt				
Dia	mm	20	25	32	36	40
Cross Section	mm²	278	478	740	961	1164
Tensile Load	kN	200	342	550	630	700
Tensile Strength	MPa	700	700	700	650	600
Nut load	kN	70	90	100	130	140
Torque	N·m	60	120	230	300	400
Shear Strength	MPa	150	150	150	150	150
E-modulus	GPa	40	40	40	40	40

GENERAL SOLID GFRP ROCKBOLT /SOIL NAIL

OCEANPOWER has the production capacity and technology of GFRP Rockbolt and FBG(Fiber Bragg Grating), combined as “Intelligent Rebar” ; Through fiber optic sensing technology, could monitoring Bridge, Slope,and Tunnel area. It’s not only meets the requirement of support, but also achieved healthy monitoring.



GFRP Rebar(Sand-Coated)

Diameter (mm)	Guranteed Tensile Strength (MPa)	Guaranteed Shear Strength (MPa)	Elasticity Modulus (GPa)
4~12	700~1000	120-150	>40
13~16	600~900		
18~25	550~800		
26~40	550~700		

Remark: More diameter as requirements.

GFRP REBAR (Thread)

GFRP Rebar, is a new construction material which use glass fiber as reinforced material, combined with resin; it's the most popular, cost-performance composite material rebar. With advantages like: High strength- Low weight, CorrosionResistance, Cuttability, Nonmagnetic, could applied in Transit Pavement Reinforcement, Metro Tunnel Shield, Underground Foundation, Coastal Development and Photo voltaic Support System etc.

Glass fiber are mainly divide into E-glass(Alkali-free) , C-glass(Medium-alkali) A-glass(High Alkali), E-CR glass(Boron& Alkali-free) and S-glass(High Strength) etc. E-glass is the most popular one.





Basalt FRP Rebar

2

BFRP is a kind of rustless electrical insulator, with non magnetic, especially with great acid resistance and alkali resistance. It could develop the durability of construction.

1

BFRP Rebar , the FRP Rebar which chose Basalt fiber as reinforcement material. Basalt fiber is produced by natural volcanic basalt rock , with environmental protection, high-low temperature resistance, heat&sound insulating, better performance (GFRP) property.

3

Applications :
Dock, Dam, mining roadway support, reinforcement construction.
Concrete road bridge reinforcement, Corrosion-resistance construction, NMR/CT house(Hospital)
Optical cable, cable core reinforcement, TV telecom transmission tower, transformer substation etc.

Mechanical Properties of BFRP Rebar

Tensile Strength(MPa)		≥750
Tensile M-modulus(MPa)		≥4.0x10 ³
Elongation(%)		≥1.8
Coefficient of thermal expansion (x 10 ⁻⁶ /℃)	Longitudinal	9~12
	Horizontal	21~22
Alkali Resistance (Strength Retention)(%)		≥85
(4 π x10 ⁻⁸ SI)		≤5x10 ⁻⁷

Remark: Reference to standard JT/T776.4-2010



Type	CFRP Rebar
Diameter (mm)	3~40
Density (g/cm ³)	1.5~1.6
Tensile Strength (MPa)	1800~2500
E-modulus (GPa)	120~165
Elongation (%)	1.3~1.5
coefficient of thermal expansion (× 10 ⁶ /℃)	0

Carbon FRP Rebar

CFRP Rebar, is a new kind of FRP rebar, which chose Carbon fiber as reinforcement, Epoxy or Vinyl as matrix, produced through pultrusion process.

Compare with other FRP rebar, CFRP Rebar with excellent tensile strength and low weight. Reduce construction weight and meet design performance requirement

	Unit	Hollow Rockbolt		
Diameter	mm	25/12	28/12	32/15
Cross Section	mm ²	321	439	554
Pitch	mm	10	10	10
Thread	Right	Right	Right	Right
Tensile Load	kN	210	300	335
Tensile	MPa	600	600	600
Torque	Nm	80	100	150
Shear Strength	MPa	150	150	150
E-modulus	GPa	40	40	40
Thread Load	kN	70	80	90

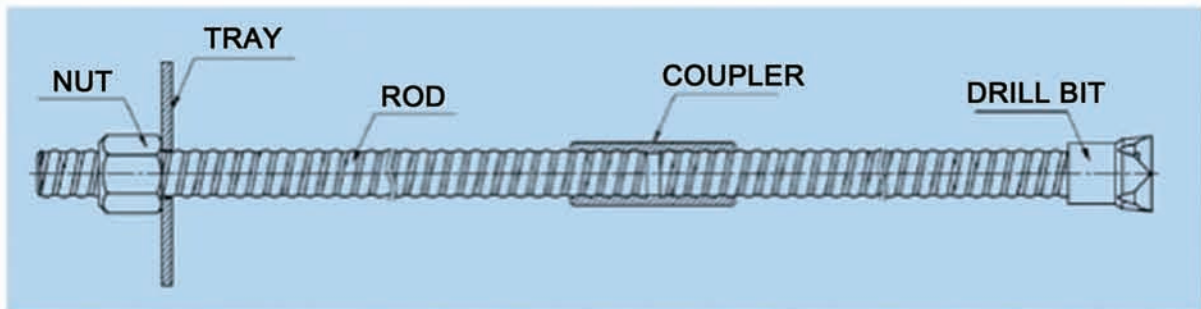
GFRP Hollow Rockbolt

GFRP hollow rockbolt is mainly used as grouting anchor, use cement&resin and other material as adhesives, could bonding main item with rockbolt, improve support ability. Like tunnel , slope stabilization and etc.



GFRP SELF-DRILLING ROCKBOLT

Self-drilling rockbolt mainly applied in incompetent bed, soft soil layer which difficult to advance pore-forming. In applications, it could fit with drill bit and assembled in drilling machine, drilling to soil directly.

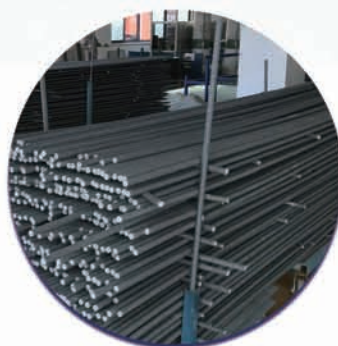


If rod length limited by transportation or applied environment, could use short rods connect by coupler, to meet design length.

	Unit	Self-drilling Rockbolt			
Diameter	mm	25/12	28/12	32/15	38/20
Cross Section	mm ²	320	430	497	704
Pitch	mm	12.7	12.7	12.7	12.7
Thread Direction		Left	Left	Left	Left
Tensile Load	kN	250	330	350	500
Tensile Strength	MPa	700	700	700	700
Torque	Nm	120	200	300	400
Shear Strength	MPa	150	150	150	150
E-modulus	GPa	40	40	40	40



GFRP
Form Tie Rod

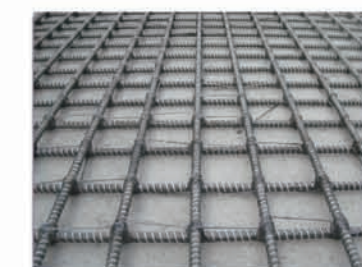
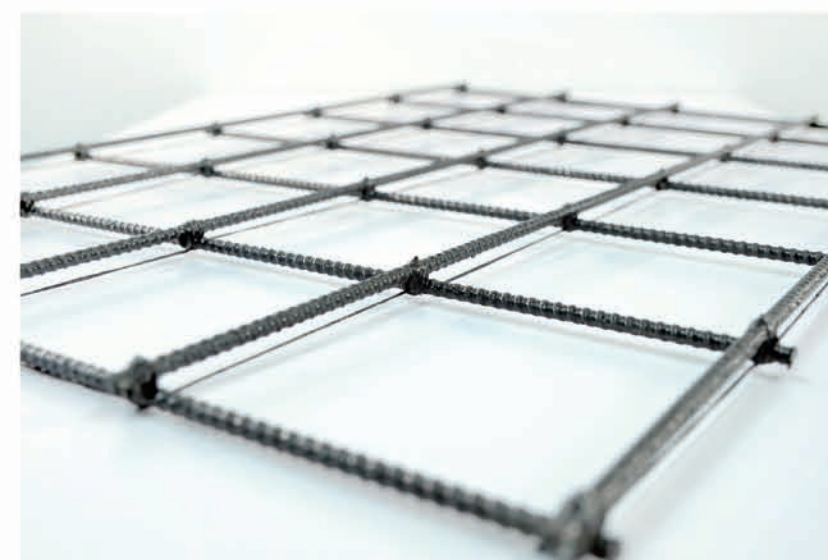


Diameter (mm)	Tensile load (kN)	Thread Breaking Load (With Holder) (kN)		Cross section (mm²)	Tensile Strength (MPa)	E-modulus (GPa)	Torque (Nm)	Weight (g/m)	Shear Strength (MPa)
		(Steel)	(FRP)						
17	150	60	/	188	800	≥45	50	390	≥120
22	250	120	/	334	750		100	700	

GFRP Mesh

	Unit	Standard Size4mm*100*100	Standard Size6mm*100*100
Rod Dia	mm	4	6
Spacing	mm	100x100	100x100
Rod Tensile Load	KN	>15	>30
Rod Tensile Strength	N/mm²	>1000	>1000
E-modulus	Gpa	>40	>40
Knot Load	KN	1<x<3	3<x<5
Standard Size	mm	1200x1800	1200x1800
Weight	g	1800	2800

G F R P





GLASS FIBER ANCHOR CABLE

The GFRP Cable bolt with it's range if products services all applications that require long bolts, like long bolting in pilot tunnels, in confined space or bolting set from parallel tunnel. It can also be used for face stabilization or slope protection, due to the easy mounting the GFRP bolt can be assembled at factory or delivered to construction site. It is therefore possible to meet customers requirements for best options.

We can supply two kind of GFRP Cable bolt, for temporary or permanent applications, for better using quality and cost control. Even GFRP Cable bolt has low weight, but also could keep stable while it's bearing load capacity.

Single Rod				
Color	Unit	Green	Blue	Black
Diameter	mm	6	6	6
Weight	g/m	60	60	60
Breaking load	kN	30	28	28
Elongation at break	%	2.1	2.1	2.1
Bending radius	m	Approx. 0.3	Approx. 0.3	Approx. 0.3
Cable Bolt				
Number of strands	pcs./bolt	6 - 90	6 - 90	6 - 90
Breanking load	kN	6 strands :180	6 strands :168	6 strands :168
		20 strands :600	20 strands :560	20 strands :560
		90 strands :2,700	90 strands :2,520	90 strands :2,520
Breaking load (Bolt head)	kN	0-800	0-800	0-800



High Performance Durability GFRP Rebar

High Performance Durability GFRP Rebar, which made of high durability resin & glass fiber, with great performance and durability, the best way to solve steel rebar corrosion; could applied in Chemical plant, Acid-base environment, Coastal environment, Bridge

Type/Dia (mm)	Tensile Load (kN)	Tensile Strength (MPa)	Shear Strength (MPa)	E-modulus (GPa)
6	36.00	1200	150.00	40.00
10	75.00	1000	150.00	40.00
14	128.00	950	150.00	40.00
20	250.20	900	150.00	40.00
25	416.30	850	150.00	

High Torque GFRP Rockbolt

We overcome the traditional difficulty of FRP rockbolt (Dowel) low torque, and invented the high torque GFRP Rockbolt. This new product could improve the supporting stability by prestressing

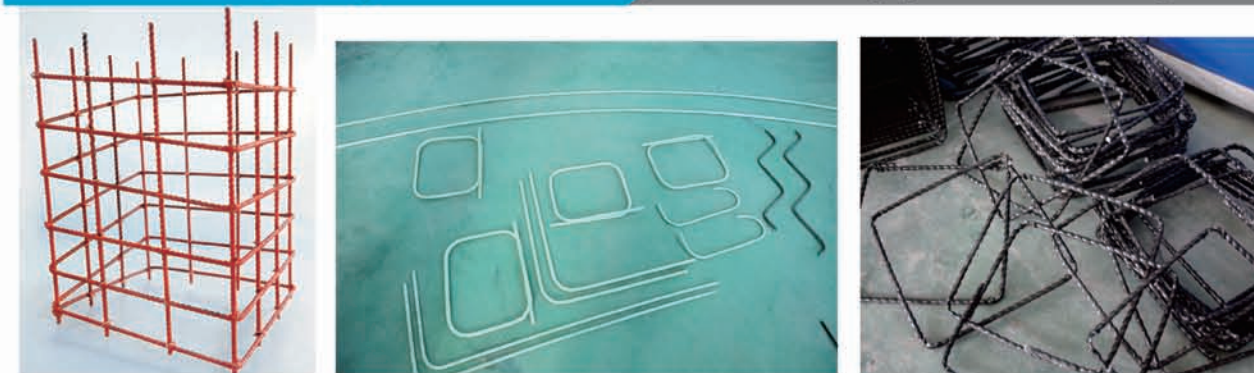


Type	Normal	HT I	HT II
Cross Section (mm ²)	340	340	340
Tensile Load (KN)	>220	>210	>210
Shear Load (KN)	>100	>100	>100
<u>Torque (Nm)</u>	<u>>80</u>	<u>>160</u>	<u>>290</u>
Nut Plate Load (KN)	>80	>80	>80

Custom Products(Special Shape)

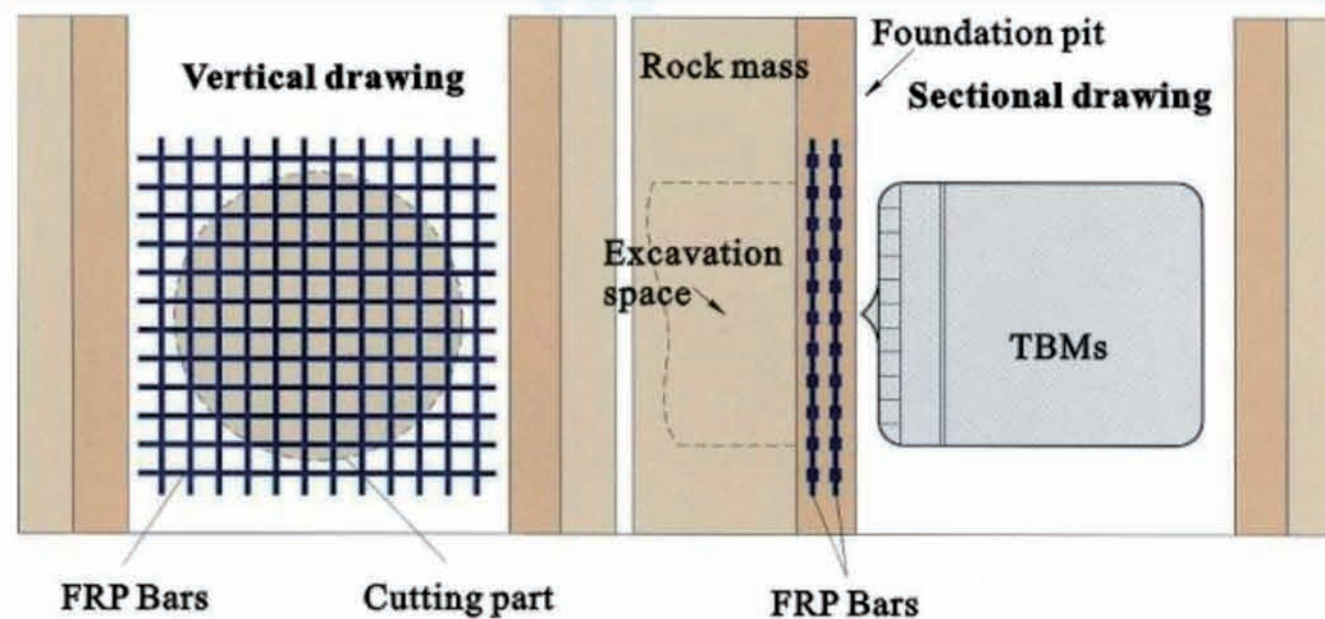


More size, shape could custom made by your drawings.





Metro TBM-Guangzhou Project-1



OCEANPOWER Developed(FRP) Metro Construction in China!

Railway Foundation-Huaian City



Metro TBM Chengdu Project

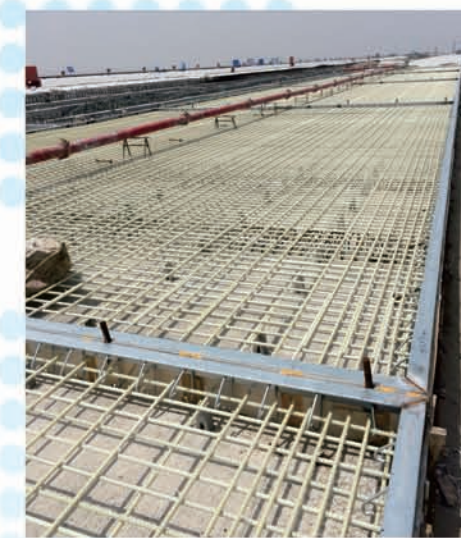




Guangzhou Huangpu Port



Shanghai Yangshan Port



Haimen City seawall



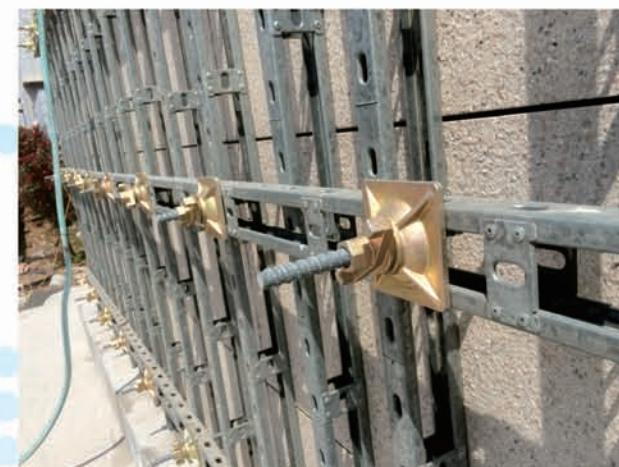
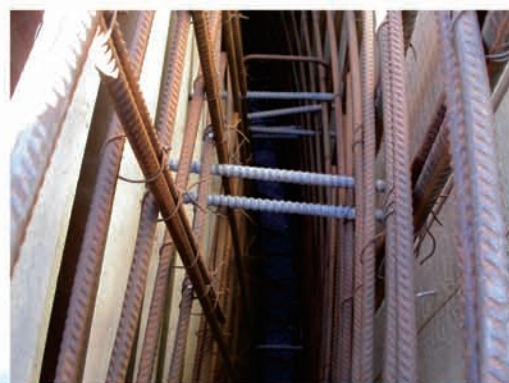


Foundation Engineering

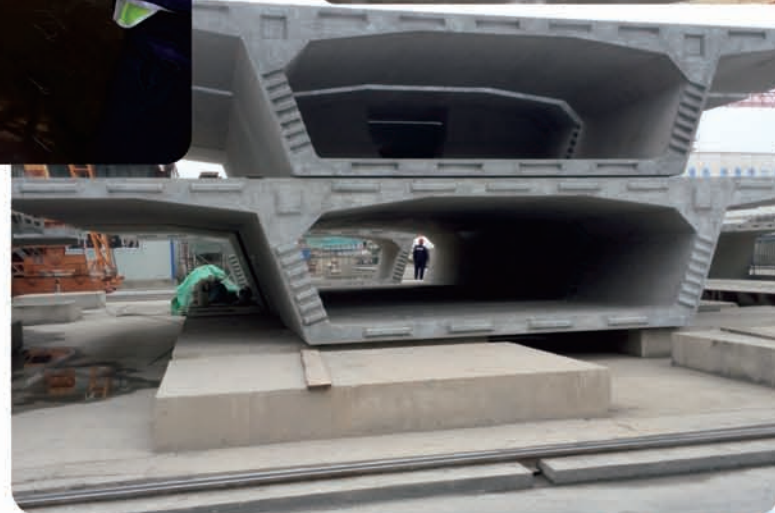
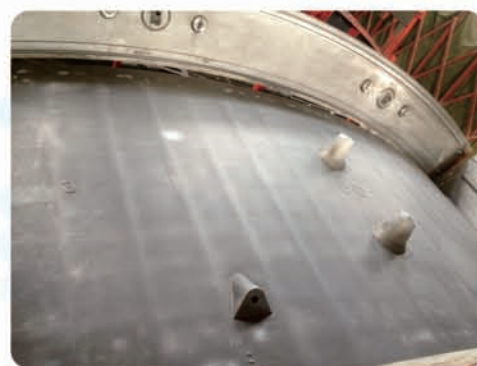
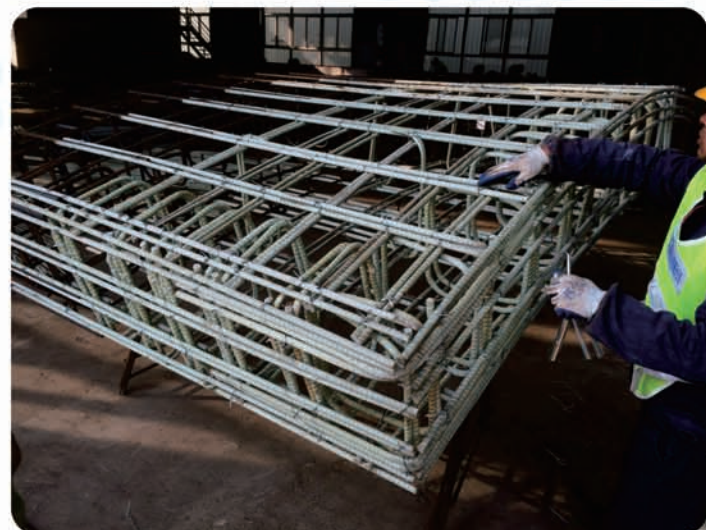


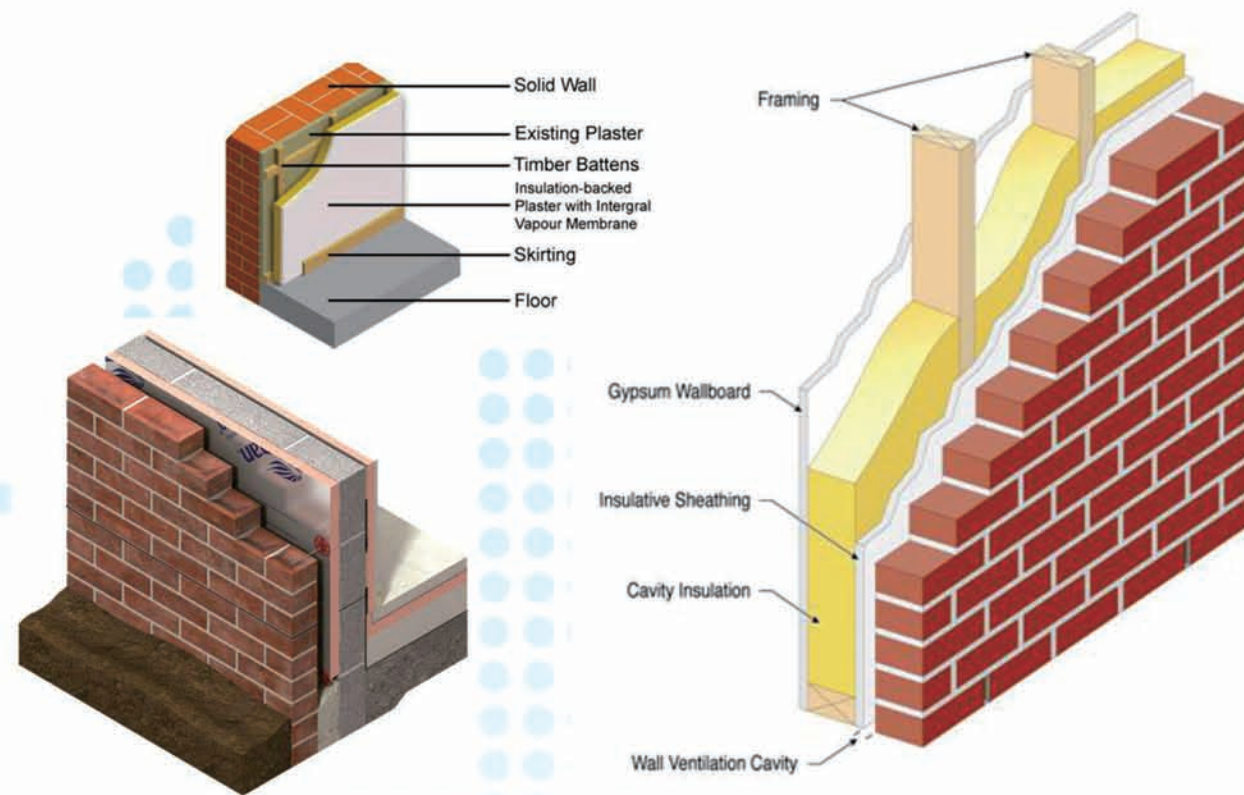
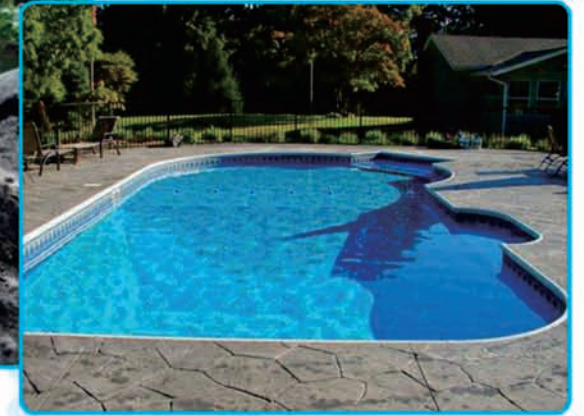
Mining Roadway Support





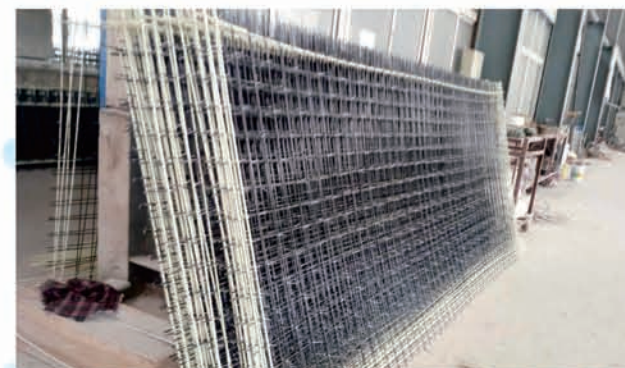
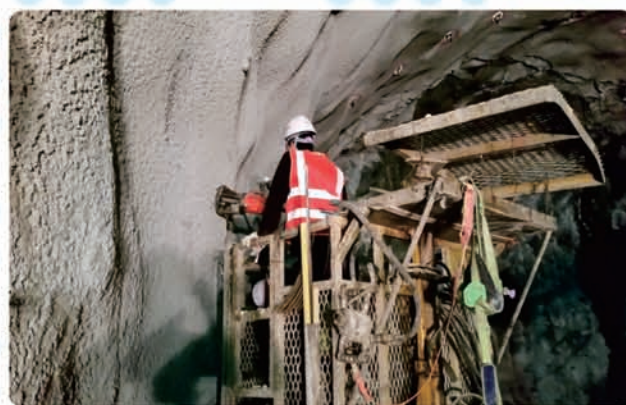
Precast Concrete Project



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Tunnel Support



Water Treatment(Metallurgy)





COOPERATIVE PARTNER

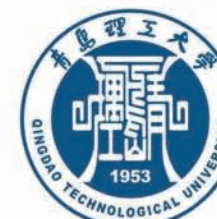
Pavement Reinforcement



中国铁建



COOPERATIVE PARTNER





FRP- A Designable Future

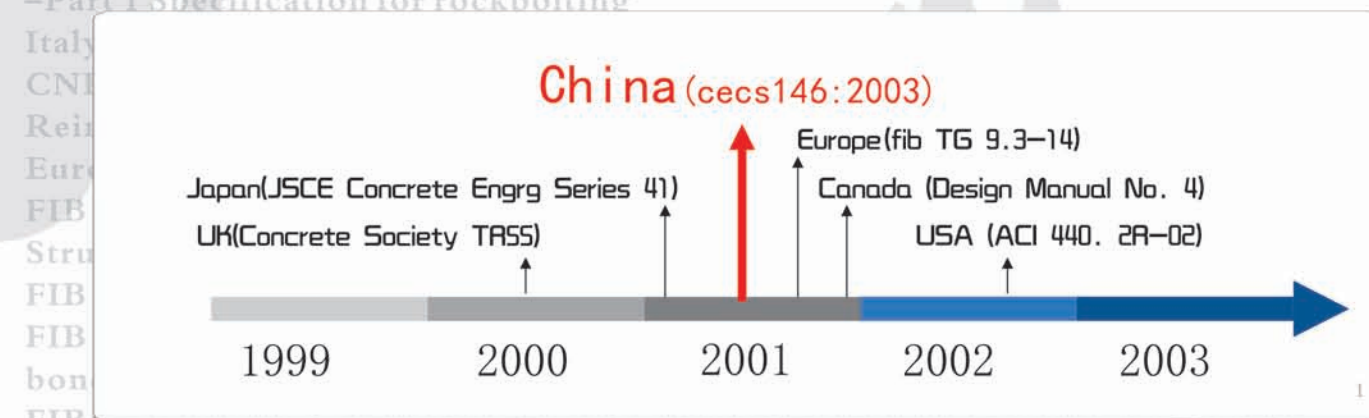
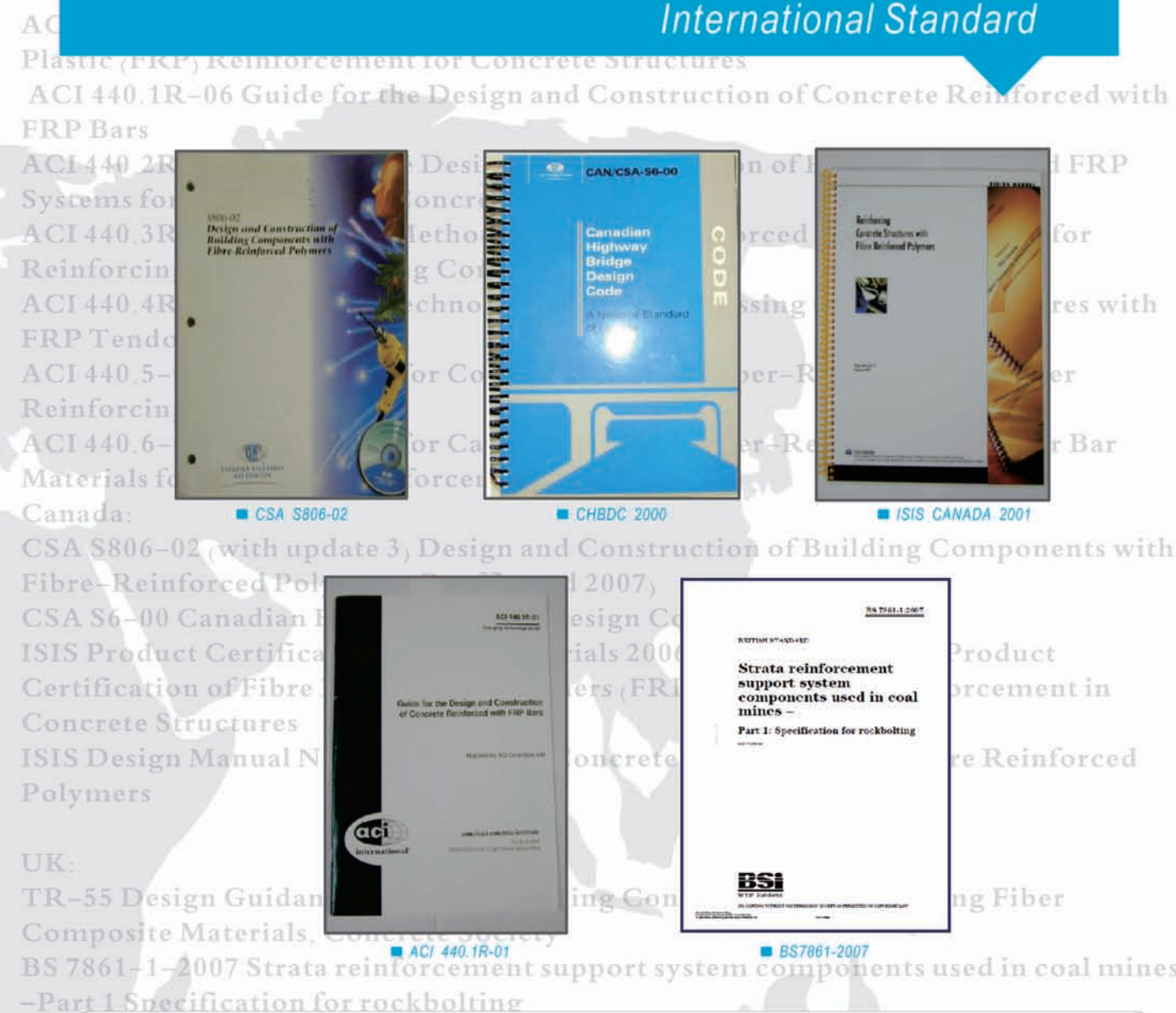


Chinese National& Industrial Standard

Standard No	Standard Name
GB50608-2010	纤维增强复合材料建设工程应用技术规范 Technical code for infrastructure application of FRP composites
GB/T 30022-2013	纤维增强复合材料筋基本力学性能试验方法 Test method for basic mechanical properties of fiber reinforced polymer bar
JG/T 406-2013	土木工程用玻璃纤维增强筋 Glass fiber reinforced plastics rebar for civil engineering
CJJ/T192-2012	盾构可切削混凝土配筋技术规程 Technical specification for shield-cutable concrete reinforcement
JG/T351-2012	纤维增强复合材料筋
MT/T 1061-2008	树脂锚杆 玻璃纤维增强塑料杆体及配件
DGJ32/TJ 162—2014	玻璃纤维增强树脂土钉基坑支护技术规程
JTT 776.4-2010	公路工程 玄武岩纤维及其制品 第4部分：玄武岩纤维复合筋
GBT 26743-2011	结构工程用纤维增强复合材料筋

American Concrete Institute--ACI

International Standard



The First FRP Standard Edit Time of Major Countries

Japan Society of Civil Engineers (JSCE), 1997, Recommendation for Design and Construction of Concrete Structures Using Continuous Fiber Reinforcing Materials